Monitoring Apache Web Server

Session 14



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Objectives

- *Describe the configuration of Apache server to improve performance*
- *Describe the configuration of caching on Apache server*
- *Describe the configuration of Apache server to track user information*

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Directive to Improve Processes

- `ServerLimit` Directive
- `StartServers` Directive
- `MinSpareServers` Directive
- `MaxSpareServers` Directive
- `MaxClients` Directive

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ServerLimit Directive

- Defines the maximum number of server processes that Apache Web server must create
- Specifies the upper limit on the number of processes that can be configured
- It is mandatory to restart Apache Web server for the changes in this directive to take effect
- Default value of the `ServerLimit` directive

MPM	Default Value
prefork	256
worker	16
perchild	8

Syntax

```
ServerLimit number
```

Snippet

```
ServerLimit 8
```

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StartServers Directive

- Defines the number of child processes that Apache Web server must create at startup
- Depending on the server load, Apache server dynamically controls the number of processes
- Configuration of this directive is optional
- Default value of `Startserver`

StartServers	Default Value
prefork	5
worker	3
perchild	2

Syntax

```
StartServers number
```

Snippet

```
StartServers 3
```

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MinSpareServers Directive

- Defines the minimum number of idle child server processes
- The MPM `prefork` supports the `MinSpareServers` directive and has a default value of 5

Syntax

```
MinSpareServers number
```

Snippet

```
MinSpareServers 7
```

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MaxSpareServers Directive

- Defines the maximum number of idle child server processes
- If the numbers of idle processes exceed the limit specified in the `MaxSpareServer` directive, then the parent processes kill the excess child processes
- MPM `prefork` supports this directive and has a default value of 10

Syntax

```
MaxSpareServers number
```

Snippet

```
MaxSpareServers 15
```

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MaxClients Directive

- Defines the maximum number of requests or connections that will be served or processed simultaneously
- The `prefork` and `worker` MPMs support this directive
- The default value is 256
- The value in the `ServerLimit` directive must be increased in order to modify the default value of the `MaxClients` directive

Syntax

```
MaxClients number
```

Snippet

```
MaxClients 40
```

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Threads

- Logical section of a program that executes independently of the other parts of the program
- Multiple threads of a program executes concurrently
- Directives that enable you to improve thread management
  - `StartThreads` Directive
  - `MinSpareThreads` Directive
  - `ThreadsPerChild` Directive
  - `ThreadLimit` Directive

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StartThreads Directive

- Defines the number of threads to be created at startup
- The `perchild` MPM supports the `StartThreads` directive
- Has a default value of 50

Syntax

```
StartThreads number
```

Snippet

```
StartThreads 8
```

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MinSpareThreads Directive

- Defines the minimum number of idle threads to process requests
- Apache server creates new threads within a child process if there are no idle threads required to process a request
- The `worker` and `mpm_netware` MPMs supports the `MinSpareThreads` directive with default values of 75 and 10 respectively

Syntax

```
MinSpareThreads number
```

Snippet

```
MinSpareThreads 10
```

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ThreadsPerChild Directive

- Defines the number of threads that can be created by each child process
- The threads are created by child processes during startup
- The `worker` MPM supports `ThreadsPerChild` directive
- Has a default value of 25

Syntax

```
ThreadsPerChild number
```

Snippet

```
ThreadsPerChild 70
```

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ThreadLimit Directive

- Defines the maximum number of threads per child process
- The `worker` and `perchild` MPMs support this directive
- Has a default value of 64 for both the MPMs

Syntax

```
ThreadLimit number
```

Snippet

```
ThreadLimit 64
```

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Performance

- Depends on the configuration of the directives
- Directives that affect Apache Web server's performance
  - `HostNameLookups` Directive
  - `AllowOverride` Directive
  - `Options` Directive

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HostNameLookups Directive

- Configures Apache Web server to log the hostname instead of the client's IP address
- The core module supports this directive
- Enabling this directive affects Apache Web server's performance as it increases the time required to complete a request

Syntax

```
HostNameLookups On|Off|Double
```

Snippet

```
HostNameLookups On
```

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AllowOverride Directive

- Enables you to include directives allowed in `.htaccess` files
- The core module supports the `AllowOverride` directive
- On enabling, Apache Web server searches for the `.htaccess` file in each directory starting at the root directory

Syntax

```
<Directory />
AllowOverride None
</Directory>
```

Directive-Type	Directives permitted
AuthConfig	Authorization directives
FileInfo	Document type directives
Indexes	Directory indexing directives
Limit	Host access control directives
Options	Directory specific directives

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Options Directive

- Supported by the core module
- Controls the server features that are available in a specific directory

Options	Functions
ExecCGI	Enables the execution of CGI scripts
FollowSymLinks	Configures the Apache Web server to follow symbolic links. A symbolic link is a special file in Linux that contains the pathname to another file
Includes	Enables Server-side includes
IncludesNOEXEC	Enables Server-side includes but disables the #exec cmd and #exec cgi command
Indexes	Returns a formatted listing of the directory when a directory does not contain a DirectoryIndex directive in that directory. The DirectoryIndex directive defines a list of files that must be searched when a client requests a directory
MultiViews	Enables content-negotiated multiviews. Content negotiation enables Apache server to select the representation of a resource based on the media type, languages, character set, and encoding information provided by the browser
SymLinksIfOwnerMatch	Enables Apache server to follow symbolic links only for the target file where the user ID of the file is the same as the link
None	Disables all additional features
All	Enables all the options except for MultiViews

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Caching and Proxing

- Caching implies storing the commonly used data at a location from where it can be accessed quickly
- Proxying allows indirect connection to a computer or a network service through a proxy server
- Caching operation is included in the `mod_cache` module
- The `mod_cache` module is split into two sub modules:
  - `mod_disk_cache`
  - `mod_mem_cache`
- Directives supported
  - `CacheRoot` Directive
  - `CacheDirLevels` Directive
  - `CacheMaxFileSize` Directive
  - `CacheMinFileSize` Directive

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CacheRoot Directive

- Defines the location for storing the cached files
- The value for `CacheRoot` directive must be specified if the `mod_disk_cache` module has been configured
- Apache server will return a configuration file processing error if there is no value specified for the `CacheRoot` directive

Syntax

```
CacheRoot directory-path
```

Snippet

```
CacheRoot /usr/bin/log
```

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CacheDirLevels Directive

- Defines the number of levels of subdirectories in the cache directory
- The default value of the `CacheDirLevels` directive is set to 3

Syntax

```
CacheDirLevels levels
```

Snippet

```
CacheDirLevels 7
```

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CacheDirLength Directive

- Defines the number of characters allowed in the subdirectory names present in the cache directory
- The default value for this directive is 2

Syntax

```
CacheDirLength length
```

Snippet

```
CacheDirLength 3
```

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CacheMaxFileSize Directive

- Defines the maximum size for a file that can be stored in the cache
- The file size must be specified in bytes

Syntax

```
CacheMaxFileSize bytes
```

Snippet

```
CacheMaxFileSize 70000
```

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CacheMinFileSize Directive

- Defines the minimum size for a file that can be stored in the cache
- The file size must be specified in bytes

Syntax

```
CacheMinFileSize bytes
```

Snippet

```
CacheMinFileSize 50
```

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Cookies

- Text file that contains user-specific information
- A Web server creates a cookie on the client machine and uses it to identify and track user information
- Since HTTP is stateless, cookies store state information about the client-server transaction
- Used to:
  - Monitor Website traffic on Apache Web server
  - Analyze Website traffic by identifying which features are used more frequently on the Web pages
  - Strengthen server security by identifying the users and tracking their activity

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Cookies

- Directives that enables you to track cookies are:
  - `CookieTracking` Directive
  - `CookieExpires` Directive
  - `CookieName` Directive
  - `CookieDomain` Directive
  - `CookieStyle` Directive

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CookieTracking Directive

- Enables to track cookies

Syntax

```
CookieTracking on|off
```

Snippet

```
CookieTracking on
```

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CookieExpires Directive

- Specifies an expiration time for the cookie created by the `usertrack` module
- The cookies will be valid only for the current browser session if the `CookieExpires` directive is not configured

Syntax

```
CookieExpires expiry-period
```

Snippet

```
CookieExpires 7200
```

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CookieName Directive

- Enables to define a name for the cookie that is used for tracking

Syntax

```
CookieName token
```

where,

`token` - Defines a valid name for the cookie

The name can include any characters such as A-Z, a-z, 0-9, “_”, and “-”

Snippet

```
CookieName Troubleshoot
```

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CookieDomain Directive

- Defines the domain for which the cookie is created or will be tracked
- If this directive is not present, no domain is included in the cookie header field

Syntax

```
CookieDomain Domain
```

where,

`Domain` - specifies the domain name

Snippet

```
CookieDomain .apachesite.com
```

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CookieStyle Directive

- Enables to specify the format of the cookie header field

Syntax

```
CookieStyle Netscape|Cookie|Cookie2|RFC2109|RFC2965
```

- where,
  - `Netscape`, `Cookie`, `Cookie2`, `RFC2109`, `RFC2965` are cookie standards
- The three formats allowed are as follows:
  - `Netscape` - original and default format
  - `Cookie` / `RFC2109`- developed after the Netscape syntax
  - `Cookie2` / `RFC2965` - current cookie syntax

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Storing User Tracking Information

- `LogFormat` directive used to create a cookie log
- To add a user tracking cookie to the standard `Logformat` directive, enter:

Syntax

```
LogFormat "%h %l %u %t \"%r\" %>s %b %{cookie}n"
```

Option	Description
%h	Specifies the remote host
%l	Specifies the remote log name
%u	Specifies the remote username
%t	Specifies the time of receiving the request
%r	Specifies the first line of the request
%>s	Specifies the status returned to the client
%b	Specifies the size of the response. The size is specified in bytes

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Storing User Tracking Information

- To create a separate cookie log file using the `CustomLog` directive, enter:

Snippet

```
CustomLog mylogs/clickstream "%{cookie}n %r %t"
```

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Summary

- A process is a program that is currently running on the computer
- The process management directives are MPM dependent, with more than one MPM implementing each directive
- A proxy server can be configured for caching documents received from the remote host
- Cookies contain user-specific information in a text file
- The `mod_usertrack` module of Apache Web server is used to implement the cookie feature

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